

Timur Cinay

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Research Interests

nitrous oxide monitoring | Bayesian inverse modeling | ocean-atmosphere gas exchange | marine nitrogen cycle

Education

Massachusetts Institute of Technology, Cambridge, MA PhD Candidate in the Program in Atmospheres, Ocean, and Climate Thesis Advisor: Dr. Andrew R. Babbín	Sep 2021 - Present
University of Rochester, Rochester, NY B.S. in Chemistry & B.S. in Environmental Science	Sep 2017 - May 2021

Publications

1. **Cinay, T.**, A. L. Ganesan, B. M. Murphy, R. L. Tunnicliffe, M. Rigby, and A. R. Babbín, Marine N₂O outgassing as El Niño wanes, *submitted*
2. **Cinay, T.**, D. Young, N. Narváez Jimenez, C. Vintimilla-Palacios, A. Pila Alonso, P. B. Krummel, W. Vizúete, and A. R. Babbín, Marine emissions and trade winds control the atmospheric nitrous oxide in the galapagos islands, *Atmospheric Chemistry and Physics*, 25(9), 4703–4718, doi: 10.5194/acp-25-4703-2025, 2025b
3. **Cinay, T.**, D. Dumit, R. J. Woosley, E. L. Boles, J. V. Kwiecinski, S. Mullen, T. J. Tamasi, M. J. Wolf, C. L. Kelly, N. M. Travis, K. L. Casciotti, and A. R. Babbín, Coincident biogenic nitrite and pH maxima arise in the upper anoxic layer in the Eastern Tropical North Pacific, *Global Biogeochemical Cycles*, 36(12), e2022GB007470, doi: 10.1029/2022GB007470, 2022

Contributed Presentations

1. **Cinay, T.**, A. L. Ganesan, N. Narváez Jimenez, D. Young, and A. R. Babbín, Atmospheric measurements reveal an El Niño fingerprint on eastern tropical Pacific nitrous oxide fluxes, in *2025 Gordon Research Conference in Chemical Oceanography*, GRC, Manchester, NH, USA, **poster**, 2025a
2. **Cinay, T.**, and A. R. Babbín, Monitoring marine nitrous oxide emissions from the oxygen minimum zones at the Galapagos Emissions Monitoring Station, in *2024 Ocean Sciences Meeting*, AGU, New Orleans, LA, USA, **poster**, 2024
3. **Cinay, T.**, D. Dumit, R. J. Woosley, E. L. Boles, J. V. Kwiecinski, S. Mullen, T. J. Tamasi, M. J. Wolf, C. L. Kelly, N. M. Travis, K. L. Casciotti, and A. R. Babbín, Coincident biogenic nitrite and pH maxima arise in the upper anoxic layer in the Eastern Tropical North Pacific, in *2022 Gordon Research Seminar in Ocean Biogeochemistry*, GRC, Barcelona, Spain, **talk**, 2023

Awards & Fellowships

MIT Libraries, The MIT Prize for Open Data	Oct 2025
MIT EAPS, The Houghton Fellowship	May 2025
MIT EAPS, The Award for Excellence in Teaching	May 2024
MIT EAPS, The Community Builder Award	May 2024
MIT School of Science, The Phillips Fellowship in Environmental Sustainability	Jan 2022
MIT, Presidential Fellowship	Sep 2021
MIT EAPS, Charney Prize	Sep 2021
University of Rochester, Annalise Kjolhede Memorial Prize for Academic Excellence in Environmental Sciences	May 2021
University of Rochester, Chemistry Department Award	May 2021

Teaching Experience

Teaching Assistant, MIT 12.004: Introduction to the Global Carbon Cycle	Sep 2025 - Dec 2025
Teaching Assistant, MIT 12.373/777: Field Oceanography	Feb 2024 - May 2024
Undergraduate TA, U. of R. CHEM231W: Chemical Instrumentation	Sep 2020 - Dec 2020
Undergraduate TA, U. of R. EESC213: Hydrology and Water Resources	Sep 2020 - Dec 2020
Undergraduate TA, U. of R. EESC103: Introduction to Environmental Science	Jan 2020 - May 2020
	Jan 2019 - May 2019

Professional Training

Graduate Teaching Certificate Program, MIT Teaching + Learning Lab	2024 - 2025
Surface Ocean Lower Atmosphere Study (SOLAS) Summer School, Mindelo, Cabo Verde	2023

Service & Outreach

Peer Review: *Global Biogeochemical Cycles*

Mentor, EAPS Application Mentorship Program, MIT	Oct 2024 - Present
Lead, EAPS Queer Alliance, MIT	Nov 2022 - May 2025
Officer, EAPS Student Advisory Committee, MIT	Oct 2021 - May 2025
Lead, Atmospheric Inverse Modeling Reading Group, MIT	Sep 2024 - Dec 2024
Co-Chair, Graduate Student Advisory Group to “Climate Science” Faculty Search	Feb 2024 - May 2024

Fieldwork Experience

Installation and Subsequent Maintenance Visits to the Galapagos Emissions Monitoring Station, Puerto Baquerizo Moreno, Ecuador	May 2023 - Present
Research/Teaching Cruise in the Western North Atlantic, Woods Hole, MA, USA	May 2024
Research Cruise in the Eastern Tropical South Pacific, Concepción - Arica, Chile	Nov 2023 - Dec 2023

Datasets

1. **Cinay, T.**, and A. R. Babbin, Nitrous oxide emissions estimates in the eastern tropical south pacific with inverse modeling, doi: 10.5281/zenodo.17373585, 2025
2. Babbin, A. R., and **T. Cinay**, Monthly atmospheric nitrous oxide (N_2O) and carbon monoxide (CO) mixing ratio measurements from San Cristobal, Galápagos Islands collected using a cavity ring-down spectrometer, doi: 10.26008/1912/bco-dmo.917743.1

Miscellaneous

Professional Memberships: American Geophysical Union (AGU)

Programming & Software: MATLAB, Python, Unix/bash, LaTeX, Adobe Illustrator.

Languages: Turkish (Fluent), English (Fluent), French (Intermediate), Spanish (Elementary)